

# Q7 Coal Assignment or

Name: Yash Ray  
Roll no: 24K-0737

Q#1..

Step#1:-

|   |
|---|
|   |
| 7 |
| 3 |

push can  
push ebn

Step#2:-

|    |
|----|
|    |
| 10 |
| 3  |

pop can  
(i) can = 7  
(ii) push 10.

Step#3:-

|            |
|------------|
| Empty<br>↓ |
|------------|

pop ebn  
pop can  
ebn = 10  
can = 3

*Ans.*

Name: Yash Raj

Roll no.: 24K-0737

Q#2:-

• data

num1 dword 4

num2 dword 5

num3 dword 6

num4 dword 7

result dword 0

• code main Proc.

mov eax, num2

cmp num, eax

Jle cluif ; move to cluif.

mov eax, num4

cmp num3, eax

Je cluif ; if (num3 == num4) cluif.

mov result, 0

mov eax, num1

addl eax, num3

addl result, eax ; result = num1 + num3

Jmp done

cluif

mov eax, num4

cmp num2, eax

JNE clu

mov eax, num2

sub eax, num1

mov result, eax

jmp done

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Roll no:- 24K0732

cli:

mov rax, 0

jmp done

done:

cmt

main ENDP

END main



Name: Yash Raj

Roll no: 24K-073

Q073-)

```
INCLUDE Irvine32.inc  
.data
```

```
Sum    dword 0  
i      dword 0  
Count  dword 5  
Array  dword 1, 2, 3, 4, 5
```

```
.code
```

```
main PROC
```

```
    mov esi, 0
```

```
l1:
```

```
    mov ecx, Count
```

```
    cmp i, ecx
```

```
    jge endloop
```

```
    mov edx, array[esi]
```

```
    cmp edx, 0
```

```
    jle ecx
```

```
    add Sum, edx
```

```
    add esi, 4
```

```
    inc i
```

```
    jmp l1
```

Namer-Yash-Raj

Roll-no:- 24K-0737

else:

mov ecx, -1

sub sum, ecx

add ecx, 4

jmp L1

exit loop:

exit

main ENDP

END main.

Namer-Yash-Raj

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Q#4:-

→ final value of An after completion of do-loop.

An = FFF6h or -10 Ans.

Name: Yash Ray

Roll no: 24K-073

Q#5c)

mov al, 0D4h ; al = 1101 0100 or D4h  
shl al, 3 ; al = 10100000 or 0A0h

mov al, 0D4h ; al = 1101 0100  
sarl al, 3 ; al = 10100000 or 0A0h  
stc ; CF = 1

mov al, 0D4h ; al = 1101 0100  
rol al, 1 ; al = 1010 1001 or 0A9h  
stc ; CF = 1

mov al, 0D4h ; al = 1101 0100  
rcr al, 3

in 1st step: CF = 1  
1110 1010 → [CF = 0]

in 2nd step:  
0111 0101

in 3rd step: [CF = 1]  
1011 0100 → 0BAh → al

My



Name- Yash Raj

Roll no- 24K-0737

Ques 6.

INCLUDE Irvine32.inc

.data

Op1 dword 5 ; assumed value

Op2 dword 6

x dword 0

y dword -1

z dword 0

.code

L1: while-loop

mov eax, Op1

cmp eax, Op2

Jg exitloop

mov eax, x

cmp Op1, eax

Jge checkanotherCondition

sub z, 10 ; else condition

sub Op1, 1

jmp L1

checkanotherCondition:

mov eax, y

cmp Op1, y

Jg else

add z, 10

dec Op1

jmp L1

Name: Ash-Raj

Roll no: 24K0737

clr:

sub 2, 10

sub opt, 1

jmp 1

ent loop:

ent

main ENDP

END main

*Am*



Name: Yash Raj

Roll no: 24K-0737

Q112.

INCLUDE Irvine32.inc

.data

val byte -11

msg byte "positive", 0

msg1 byte "negative", 0

msg2 byte "zero", 0

.code main PROC

mov ecx, val

cmp ecx, 0

lj pos

; go to positive

jl ny

; go to negative

mov ecx, offset msg2 ; the print zero

call WriteString

call crlf

jmp done

pos:

mov ecx, offset msg

call WriteString

call crlf

jmp done

ny:

mov ecx, offset msg1

call WriteString

call crlf

jmp done

done: exit main END END main

*Am*

Name: Yash Raj

Roll no: 248072

Ques.

INCLUDE Irvine32.inc

.data

var1 sword

var2 sword

result sword ?

my byte Result, 0

.code

main PROC

mov ax, var1

mov bx, var2

imul bx

movsx ecx, ax

shl ecx, 16

or ecx, ecx

mov result, ecx

mov ecx, result

mov ecx, offset my

call WriteString

call crlf

call WriteInt

exit

main ENDP

END main



Name: Yash Raj

Roll no: 24K-0737

Q29:-

INCLUDE Irvine32.inc

.data

array word 10, 60, 20, 33, 72, 89, 45, 65, 72, 18

sample word 50

Arraysize word ?

index word 0

sum word 0

.code

main PROC

mov ax, SIZEOF array

mov bx, SIZEOF sample

xor cx, cx

div bx

mov Arraysize, ax

While:

mov ecx, index

cmp ax, Arraysize

mov cx, index

mov ax, sample

cmp arr[ecx], ax

JA donothing

mov bx, arr[ecx]



Namer Yash-Raj

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add sum, bx

inc inc bx

jmp while

done thing:

inc inc bx

jmp while

exit loop:

exit

main ENDP

END main

Name: Yash Ray

Roll no: 24K07

SHLO: Part (i).

INCLUDE Irvine32.inc

• data

num dword ?

• code

main PROC

call ReadInt

mov ebx, eax

mov ecx, eax

shl ebx, 4

shl ecx, 3

shl eax, 1

add ebx, ecx

add eax, ebx

mov num, eax

cmt

main ENDP

END main

Part (ii): No change in the result  
because i am taking input it  
or writes over eax, (user defined)  
value.

